A two-coset criterion for chiral hypertopes

Human Compression project

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Abstract

We derive the chirality criterion of Zhang and Leemans, Theorem 1.1 of their November 2025 author manuscript, from chamber fibers and an explicit reflection of a chamber graph. The finite cardinality test is written as a statement about two cosets, which is also meaningful for infinite groups. We prove residual connectedness directly, including the cyclic rank-two residues, and distinguish the designated group from the full type-preserving automorphism group until that distinction is resolved. The target is this criterion, not a replacement of the source's implementation or every ancillary assertion.

1 The marking and the exact criterion

Let $I = \{0, \dots, r-1\}$, where $r \geq 3$, and let

$$G = \langle \alpha_1, \dots, \alpha_{r-1} \rangle, \qquad \alpha_0 = 1, \qquad a_{ij} = \alpha_i^{-1}\alpha_j, \qquad P_S = \langle a_{ij} : i, j \in S \rangle.$$

We use the C^+ marking of Zhang–Leemans: the specified nonidentity generators are independent, and

$$P_S \cap P_T = P_{S\cap T} \qquad (|S|, |T| \geq 2). \tag{1}$$

Singleton and empty supports generate the trivial group. In particular the α_i are distinct. Put $H_i = P_{I\setminus\{i\}}$ and let Γ be their coset incidence geometry. The word geometry is a hypothesis: every flag extends to a chamber. Our left cosets correspond to the source's right cosets by $gH_i \mapsto H_i g^{-1}$; this bijection preserves nonempty intersection and converts the acting element to its inverse.

Repeated use of (1) gives

$$H_J := \bigcap_{j\in J} H_j = P_{I\setminus J}. \tag{2}$$

Indeed, intersect successively while the remaining support has at least two elements; once its size is one, the group is already trivial. Thus $H_J = 1$ whenever $|J| \geq r-1$. The panel-stabilizer condition listed separately in source Theorem 1.1 is consequently automatic under this marking hypothesis.

Theorem 1.1. *The geometry Γ is a chiral hypertope if and only if the following conditions hold:*

(i) for some $k \in I$, G is transitive on the chambers of the truncation omitting type k;

(ii) for that k, the normalized extension set is exactly two cosets:

$$\bigcap_{j\neq k} H_j H_k = H_k \,\dot\cup\, xH_k \quad \text{with } x \notin H_k; \tag{3}$$

(iii) no automorphism of G sends every α_i to α_i^{-1}.

When these conditions hold, condition (i) and the corresponding two-coset condition hold for every k. If the groups are finite, (ii) is equivalent to the product-size condition in Zhang–Leemans, Theorem 1.1. For infinite groups the exact coset condition, not that cardinal equation, is the statement proved here.

2 Neighbors and proper flags

Let $C = \{H_i : i \in I\}$ be the base chamber. For $i \in I$ define

$$D_i = (C \setminus \{H_i\}) \cup \{a_{ij}H_i\}, \qquad j \neq i. \tag{4}$$

The chosen j does not matter: $a_{ij}^{-1}a_{ik} = a_{jk} \in H_i$ when $j, k \neq i$. This is a chamber. To check its modified object against H_l, choose a representative a_{ij} with $j \neq i, l$; this is possible because $r \geq 3$, and the representative lies in H_l. Also $a_{ij} \notin H_i$: otherwise (1) applied to $\{i, j\}$ and $I \setminus \{i\}$ would put a_{ij} in $P_{\{j\}} = 1$, contrary to the distinctness of α_i, α_j. Thus C and D_i are distinct and differ only at type i.

The key transport identity is

$$a_{ji}D_i = D_j. \tag{5}$$

At type i the multiplier cancels a_{ij}; at type j it creates the modified coset of D_j. At every other type l, both i, j differ from l, so $a_{ji} \in H_l$ and the coset remains H_l. This verifies the identity at every coordinate.

Assume (i)–(ii). The extension-fiber theorem in the common paper, together with (2), gives exactly two free chamber orbits. They are represented by C and D_k: no element of G can carry C to D_k, because it would fix their panel, whose stabilizer is trivial. Formula (5) puts all D_i in the second orbit.

Every proper flag is in the orbit of a flag of C. To see this, extend it to a chamber and move that chamber to C or D_k. In the second case choose a type i missing from the flag and move D_k to D_i by (5). Its objects of the flag's types then coincide with those of C. Hence G is transitive on every proper flag type.

Fix now the panel of C missing i. It has the two completions C, D_i. Any other completion is in one of their two free orbits. An element carrying the appropriate representative to that completion fixes the whole panel and must be the identity. The panel therefore has exactly those two completions. Transitivity on panels proves thinness everywhere. This argument counts completions within each orbit and is valid for infinite G; it does not infer a panel size by dividing infinite cardinalities.

3 Every residue is connected

Normalize a flag of type J to $\{H_j : j \in J\}$ and set $S = I \setminus J$. For $i \in S$, transitivity on flags of type $J \cup \{i\}$ shows that the type-i objects of the residue are exactly

$$H_J/(H_J \cap H_i).$$

If $|S| \geq 3$, this assertion uses only proper flag transitivity. The same is true when $|S| = 2$, because $|J \cup \{i\}| = r - 1$. The internal coset incidence system has all these objects and some of the residue's edges; it is a spanning subgraph, not automatically an incidence isomorphism.

When $|S| \geq 3$, its defining subgroups are $P_{S\setminus\{i\}}$. They generate $P_S = H_J$, since each generator a_{uv} belongs to $P_{S\setminus\{i\}}$ for an i different from u, v. For completeness, generation implies connectivity as follows. The objects represented by the identity are mutually incident. Multiplying their whole chamber by an element of one defining subgroup leaves that subgroup's object fixed, so the two chambers lie in the same connected component. Induction on a word in the defining subgroups brings every group translate into this component. These translates contain every object. The larger actual residue graph is therefore connected too.

Suppose $S = \{u, v\}$. Then $H_J = \langle a_{uv} \rangle$ is cyclic and both vertex stabilizers are trivial. The residue has two copies of this cyclic group. It contains the matching edges

$$h_{\text{left}} \; -\!- \; h_{\text{right}}, \qquad h_{\text{left}} \; -\!- \; (ha_{uv}^{-1})_{\text{right}}.$$

The first edges come from hC. The second come from hD_v, since D_v has its type-v object represented by $a_{vu} = a_{uv}^{-1}$. Two successive matching steps move the cyclic coordinate by a_{uv} or its inverse, so these

edges connect all vertices. They form a cycle in the finite case and a doubly infinite path in the infinite case. Any additional incidence edges preserve connectivity. This proves residual connectedness for every flag and completes the hypertope part of the sufficient direction.

4 The full automorphism group and the converse

Take D_0 as the negative base chamber. Equation (5) gives

$$D_i = \alpha_i^{-1} D_0.$$

Label the two free chamber orbits by $(g, +) = gC$ and $(g, -) = gD_0$. Thinness gives exactly the adjacency formula

$$(g, +) \sim_i (g\alpha_i^{-1}, -)$$

used by the common reflection theorem. That theorem proves that (iii) is equivalent to chirality and identifies the full type-preserving automorphism group, rather than assuming it equals the designated group G.

For necessity, suppose Γ is a chiral hypertope, with chirality defined using its full type-preserving automorphism group. We first show directly that the designated G has the two chamber orbits required above. The chambers D_i are the unique i-neighbors of C by thinness. Let $\mathcal{C}$ be the union of the G-orbits of C and D_0. Every neighbor of gC is gD_i, in the second orbit. Conversely every i-neighbor of gD_0 is $g\alpha_i C$: translate the i-edge from C to $D_i = \alpha_i^{-1} D_0$ by α_i. Thus $\mathcal{C}$ is closed under adjacency. Residual connectedness implies chamber connectedness by the common propagation lemma, so $\mathcal{C}$ is the entire chamber set.

The two orbits are distinct. If $gC = D_0$, then g fixes their common panel and lies in $H_{I\setminus\{0\}} = 1$, which is impossible. The stabilizer of a chamber is also trivial. The proper-flag argument of Section 2 now gives transitivity on every truncation. Normalizing a panel, its exactly two completions give precisely two cosets by the extension-fiber theorem, proving (i)–(ii) for every k. Finally the common reflection theorem would make the full geometry regular if (iii) failed, contrary to chirality. This proves necessity without presupposing that the designated action is the full chiral action.

5 Source scope and attribution

The exact principal target is Theorem 1.1 of W.-J. Zhang and D. Leemans, *Testing chirality on hypertopes*, arXiv:2511.14562v1, the ten-page November 2025 manuscript. All ten acquired pages have been read for this edition. The source states the cardinality test without an explicit finiteness restriction; the two-coset formulation above records the necessary infinite-group qualification. Condition (ii) concerning panel stabilizers in the source is retained as a proved consequence of its C^+ hypothesis.

The orbit normalization also explains the source's Theorem 3.2, but this edition does not enlarge the admitted principal target to a replacement of every ancillary result or the three Magma functions printed in Section 5. Those functions and their software dependencies are not required by the uniform written proof here. The source's passage from a marked inversion to a geometric reflection cites Fernandes–Leemans–Weiss (2016), Theorem 8.2; the common paper supplies the actual chamber map and its geometric reconstruction. The residue argument reuses the earlier reviewed proof bank's distinction between an internal coset graph and the larger actual residue, including the second matching in rank two. No finite experiment is used to establish the quantified theorem, and no historical novelty is asserted for the criterion.